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Yannick Becker, Ralf Korn, Pascal Halffmann

Lecture 3A: Continuous-Time Portfolio Optimization – Some Practical Issues

Ralf Korn (RPTU Kaiserslautern-Landau, Fraunhofer ITWM)

RPTU Rheinland-Pfälzische Technische Universität Kaiserslautern Landau

Fraunhofer ITWM

Practical Issues:

- We ignored transaction costs
- We traded at every time instant

  ⇨ How can this work in reality?

- We assumed that all parameters are known

  ⇨ How to get them?

How to apply continuous-time portfolio optimization in reality?

So far, all worked well … but:

Typical optimal strategy:

Always invest 60% of your wealth in the stock and 40% in the money market account.

⇨ Structurally very clear, simple and easy to understand

⇨ However: one has to trade at **every** (!!!) time instant to keep the fractions constant

Even worse: The trading strategy has infinite variation

$$\varphi_1(t) = \frac{\pi X(t)}{S_1(t)} = \frac{\pi x}{S_1(0)} e^{\left(r+\pi(b-r)-\frac{1}{2}\pi^2\sigma^2\right)t+\pi\sigma W(t)-\left(b-\frac{1}{2}\sigma^2\right)t-\sigma W(t)}$$
$$= \frac{\pi x}{S_1(0)} e^{\left(r+\pi(b-r)-\frac{1}{2}(\pi^2-1)\sigma^2\right)t+(\pi-1)\sigma W(t)}$$

Hence: In the presence of any **reasonable form** of transaction costs, we will **immediately** be ruined (???)!!!

WHAT TO DO?

Do we really have to trade so often?

R. Korn (2025) Can continuous-time portfolio optimization really be applied?

=>

Natural strategy (for d=1)**:**

- Take the optimal portfolio process $\pi^*(t)$ as an orientation
- Act only if your actual fractions $\pi(t)$ are far away from $\pi^*(t)$, i.e. :
- Determine intervals I(t):= $\left[\pi_{low}(t), \pi_{up}(t)\right]$ that contain $\pi^*(t)$.
- As long as we have $\pi_{low}(t) < \pi(t) < \pi_{up}(t)$, do nothing.
- As soon as we have $\pi(t) \in \{\pi_{low}(t), \pi_{up}(t)\}$, perform the natural action such that we get back to $\pi(t) = \pi^*(t)$,
- i.e. sell or buy the appropriate amount of shares where the costs are paid from the money market account (in case of buying) and the gains are put into the money market account (in case of selling).

Do we really have to trade so often? – 2

R. Korn (2025) Can continuous-time portfolio optimization really be applied?

=>

Natural problem: How to determine such an interval $\left[\pi_{low}(t), \pi_{up}(t)\right]$?

A very hard problem:

- Already for d=1, only approximate solutions exist (see e.g. K. (1998))
- Or solutions for specialized problems such as life-time consumption problems (see e.g. Davis, Norman (1988))
- For d > 1: What is the generalization of an interval?
- Solutions for „strange problems“ exist (such as in Morton, Pliska (1995), see also Atkinson, Pliska, Wilmott (1995) or K. (2025) for a modified problem)

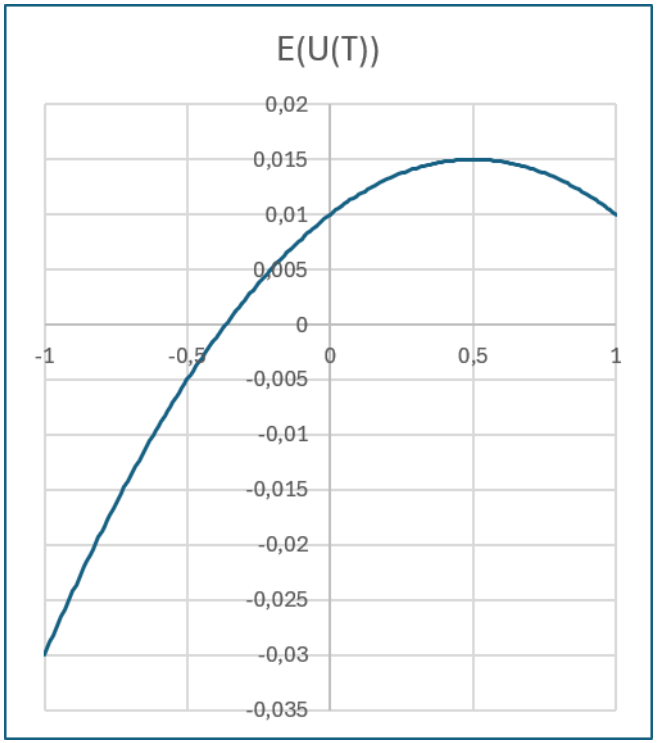
Surprising help …

Do we really have to trade so often? - 3

L.C.G. Rogers (2001) The relaxed investor

A robustness result: Expected log-utility as a function of π

- Expected utility function is flat close to the optimum
- It is steep far away from the optimum



Hence:

If one **starts in the optimal portfolio process** and does not (!!!) adjust the portfolio holdings, then this does not lead to a substantial loss of expected utility.

If one **starts far from the optimal portfolio process** and does not adjust the portfolio holdings, then this does lead to a substantial change of expected utility.

Do we really have to trade so often? – 4

L.C.G. Rogers (2001) The relaxed investor

Standard setting: *d*=1, constant coefficients, $U(x) = \frac{1}{\gamma}x^{\gamma}, \gamma < 1, \gamma \neq 0.$

Basic idea:

- The relaxed investor (RI) trades only at fixed times i=0,…,N, t_i=$i\Delta$, $\Delta = T/N$.
- RI always rebalances to $\pi(i\Delta) = p$ for some constant 0 < p < 1.
- Look at the efficiency $\theta(\Delta)$ of this strategy, i.e. the fraction of money that the continuous-time investor following the optimal portfolio process needs as initial capital to obtain the same expected utility after time Δ as the relaxed investor.

Surprising (?) results (Rogers (2001)):

By an expansion in Δ of the opt. expected utility and opt. strategy p^* for the RI:

- The efficiency is typically **very close to 1** (i.e. $\theta(\Delta) \approx 0.99$ for realistic param.)
- The optimal p^* is very close to π^* (equal in order zero)
- These assertions hold for quite large values of Δ, such as $\Delta = 1$

Do we really have to trade so often? – 5

L.C.G. Rogers (2001) The relaxed investor

Standard setting: *d*=1, constant coefficients, $U(x) = \frac{1}{\gamma}x^{\gamma}, \gamma < 1, \gamma \neq 0.$

What does this mean:

- It is **important** to know the continuous-time **optimal portfolio process** π^*.
- It is **enough** to adjust the holdings **at least annually**

  (as long as one believes in constant coefficients !)

In particular:

A **rebalancing every quarter** (or even annually) is a **sufficiently good approximation** of optimal continuous trading.

This will even be **emphasized** when taking into account the uncertainty of the estimation of the market coefficients

Still: The theoretical justification needs to be generalized to the d-stock case.

How to get the parameters?

Rogers (2001) "*Parameter uncertainty has a much bigger effect than trading frequency*"

Let us see …

Simplest case: d=1, constant coeff. b, σ, observations $S_1(t_i)$, i=0,…,N, $t_i = i\Delta$, $\Delta = T/N$.

$$\Rightarrow \quad Z_i = \frac{1}{\Delta}\left(ln\left(S_1(t_i)\right) - ln\left(S_1(t_{i-1})\right)\right) = b - \frac{1}{2}\sigma^2 + \frac{\sigma}{\Delta}\left(W(i\Delta) - W((i-1)\Delta)\right)$$

$$\sim N\left(b - \frac{1}{2}\sigma^2, \frac{\sigma^2}{\Delta}\right) \text{ are i.i.d.}$$

With $\mu := b - \frac{1}{2}\sigma^2$ we obtain the ML-estimators:

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} Z_i, \qquad \hat{\sigma}^2 = \Delta\frac{1}{N}\sum_{i=1}^{N}(Z_i - \bar{Z}_N)^2$$

Note: We have

(1) $$\hat{\mu} = b - \frac{1}{2}\sigma^2 + \frac{\sigma}{T}W(T) \sim N\left(b - \frac{1}{2}\sigma^2, \frac{\sigma^2}{T}\right),$$

i.e. the ML-estimator is independent of N and of Δ, only the last observation counts.

Thus, to make the variance of the estimator small, we only have to increase T …

How to get the parameters? - 2

Rogers (2001) "*Parameter uncertainty has a much bigger effect than trading frequency*"

Let us see …

Note: For the ML-estimator of the variance we get:

(2) $$\hat{\sigma}^2 = \Delta \frac{1}{N} \sum_{i=1}^{N} (Z_i - \bar{Z}_N)^2 = \sigma^2 \frac{1}{N} \sum_{i=1}^{N} \left(\widetilde{W}_i - \frac{1}{N} \sum_{j=1}^{N} \widetilde{W}_j \right)^2$$

with $\widetilde{W}_i = (W(i\Delta) - W((i-1)\Delta))/\sqrt{\Delta} \sim N(0,1)$ which are all independent,

i.e. the ML-estimator for the variance is independent of *b* and of the stepsize Δ.

However, it gets more accurate for an increasing number of *N* as the factor multiplying σ^2 in the second representation tends to 1 a.s..

How to judge our results?

An example with simulated data from Desmettre, Korn (2018):

Generated 1000 log-increments with $b = 0{,}05,\ \sigma = 0.2$ (i.e. μ = 0,03) for three different discretizations: daily, monthly, annually

Discretiz. Δ	$\hat{\mu}$	Conf. Int. left	Conf. Int.right	$\hat{\sigma}^2$	Conf. Int left	Conf. Int. right
1/365	0,0069	-0,23478	0,24854	0,0382	0,03504	0,04176
1/12	0,0258	-0,01810	0,06963	0,0382	0,03504	0,04176
1	0,0288	0,01614	0,04144	0,0382	0,03504	0,04176

- Note how the estimators exactly perform how we expected !!!
- Note in particular that for a 1000 year horizon, the length for the confidence interval for μ is **still nearly bigger** than μ itself !!!

Finally: Discuss why this is not so bad for the financial market!

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